Supplementary Information

Coupled Condensation–Melting Heat Transfer and Thermal Design Limits of a Steam-Assisted Pipeline Cleaning Crawler: A Probabilistic Modeling Study

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S1. Thermal sweep and Monte Carlo behavior (first suite)

Aerogel (mm)	Atmospheric, 45 min	Atmospheric, 90 min	Evacuated, 45 min	Evacuated, 90 min
5	0.1% (0.0–0.7%)	0%	51.9% (48.4–55.3%)	0%
10	19.1% (16.5–22.0%)	0%	92.4% (90.3–94.0%)	12%
15	62.4% (59.0–65.7%)	0%	97.1% (95.7–98.1%)	35%
20	86.5% (84.0–88.7%)	1%	98.6% (97.6–99.2%)	50%
30	96.2% (94.7–97.4%)	16%	98.9% (97.9–99.4%)	69%
40	99.1% (98.2–99.6%)	40%	99.6% (98.9–99.9%)	80%
50	99.2% (98.4–99.7%)	60%	99.6% (98.9–99.9%)	80%
60	100.0% (99.5–100.0%)	72%	98.9% (97.9–99.4%)	79%

Table S1. P(pack ≤ 55 °C) against aerogel thickness, 45 min (Wilson 95% interval) and 90 min.

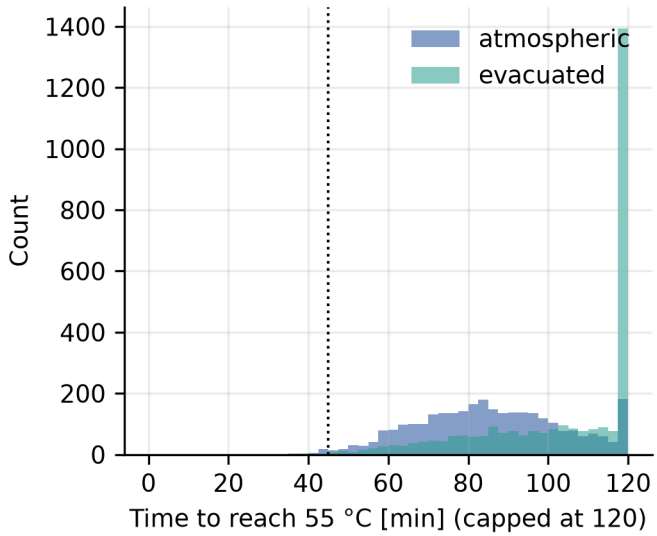


Fig. S1. Time to reach 55 °C for atmospheric and evacuated aerogel, 40 mm (capped at 120 min).

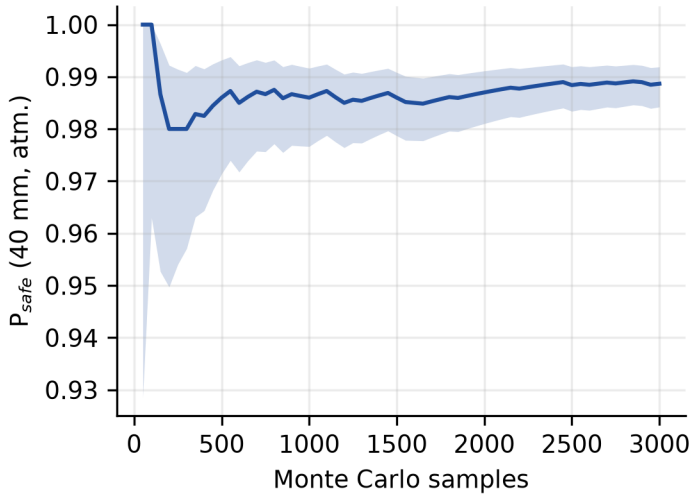


Fig. S2. Convergence of the Monte Carlo estimate of P(pack ≤ 55 °C) with the number of draws (40 mm, atmospheric); band: Wilson 95% interval.

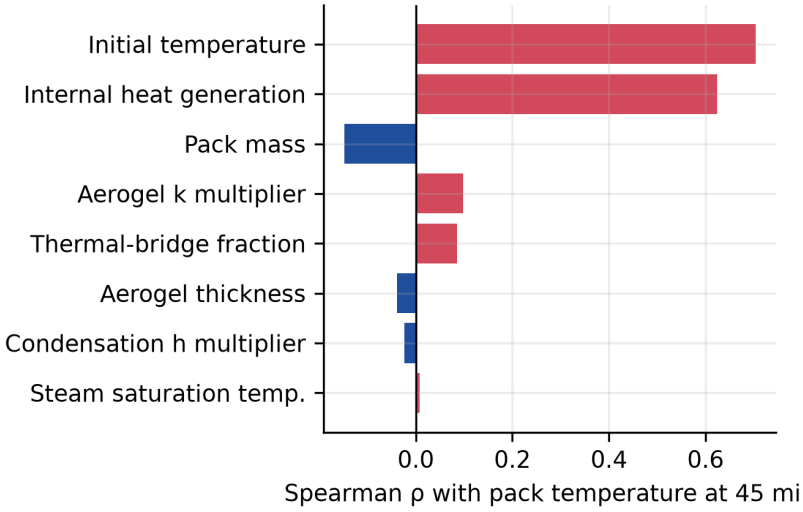


Fig. S3. Spearman rank correlation of each input with the 45 min pack temperature (first suite; global Sobol indices are in Fig. 9 of the main text).

S2. Benchmarks: all cases

h (W/m²·K)	Thickness (mm)	Ablation formula	Ablation solver	Error (%)	No-drainage formula	No-drainage solver	Error (%)
1,000	3	4.91	4.92	−0.2	54.0	65.7	−17.9
1,000	6	9.81	9.83	−0.2	206.0	251.1	−17.9
1,000	12	19.62	19.65	−0.2	804.4	980.5	−18.0
3,000	3	1.64	1.66	−1.3	50.7	61.8	−18.0
3,000	6	3.27	3.31	−1.3	199.5	243.1	−18.0
3,000	12	6.54	6.63	−1.3	791.4	964.6	−18.0
10,000	3	0.49	0.51	−4.6	49.5	60.4	−18.0
10,000	6	0.98	1.03	−4.6	197.2	240.4	−18.0
10,000	12	1.96	2.06	−4.6	786.8	959.0	−18.0

Table S2. Melt time (s) from the formulas and from the enthalpy-method solver (equal properties in both phases, T0 = 25 °C, Tm = 58 °C, steam at 198 °C, no heat-coupling loss). Errors are formula minus solver, in percent of the solver.

Cell size (µm)	Melt time (s)
16	3.650
8	3.424
4	3.313
2	3.311

Table S3. Ablation melt time of the enthalpy solver against cell size (h = 3 kW/m²·K, 6 mm).

S3. Global sensitivity: Sobol indices with 95% bootstrap intervals

Input	First order	Total
aerogel k multiplier	0.02 (−0.07 to 0.14)	0.02 (0.02 to 0.02)
aerogel thickness	0.00 (−0.04 to 0.04)	0.00 (0.00 to 0.00)
thermal-bridge fraction	0.01 (−0.06 to 0.07)	0.01 (0.01 to 0.01)
internal heat generation	0.46 (−0.18 to 1.10)	0.48 (0.42 to 0.55)
pack mass	0.02 (−0.13 to 0.16)	0.03 (0.02 to 0.03)
initial temperature	0.49 (−0.02 to 1.20)	0.49 (0.43 to 0.55)
steam temperature	0.00 (−0.01 to 0.01)	0.00 (0.00 to 0.00)
condensation multiplier	0.00 (0.00 to 0.00)	0.00 (0.00 to 0.00)

Table S4. Sobol indices of the pack temperature at 45 min (N = 512 base samples).

Input	First order	Total
deposit thickness	0.24 (0.19 to 0.28)	0.24 (0.23 to 0.24)
melt point	0.01 (0.00 to 0.02)	0.01 (0.01 to 0.01)
deposit density	0.00 (0.00 to 0.01)	0.00 (0.00 to 0.00)
latent heat	0.01 (0.00 to 0.02)	0.01 (0.01 to 0.01)
pipe temperature	0.00 (0.00 to 0.01)	0.00 (0.00 to 0.00)
heat-coupling fraction	0.11 (0.07 to 0.14)	0.11 (0.10 to 0.11)
non-condensable factor	0.00 (0.00 to 0.01)	0.00 (0.00 to 0.00)
contact length	0.07 (0.05 to 0.10)	0.07 (0.07 to 0.07)
steam temperature	0.00 (0.00 to 0.00)	0.00 (0.00 to 0.00)
condensation multiplier	0.00 (0.00 to 0.01)	0.00 (0.00 to 0.00)
retained film	0.55 (0.49 to 0.62)	0.55 (0.54 to 0.57)

Table S4. Sobol indices of the log10 range, retained film lognormal (N = 8,192 base samples).

Input	First order	Total
deposit thickness	0.45 (0.30 to 0.61)	0.48 (0.47 to 0.50)
melt point	0.02 (−0.01 to 0.06)	0.02 (0.02 to 0.02)
deposit density	0.01 (−0.01 to 0.02)	0.00 (0.00 to 0.00)
latent heat	0.02 (−0.02 to 0.06)	0.03 (0.02 to 0.03)
pipe temperature	0.00 (−0.01 to 0.02)	0.00 (0.00 to 0.00)
heat-coupling fraction	0.22 (0.11 to 0.32)	0.23 (0.23 to 0.24)
non-condensable factor	0.06 (−0.01 to 0.11)	0.08 (0.08 to 0.08)
contact length	0.14 (0.06 to 0.22)	0.16 (0.15 to 0.16)
steam temperature	0.00 (0.00 to 0.01)	0.00 (0.00 to 0.00)
condensation multiplier	0.04 (−0.01 to 0.09)	0.05 (0.04 to 0.05)

Table S4. Sobol indices of the log10 range, instant removal (N = 8,192 base samples).

Input	First order	Total
deposit thickness	0.65 (0.61 to 0.68)	0.65 (0.63 to 0.67)
melt point	0.01 (0.01 to 0.02)	0.01 (0.01 to 0.01)

Input	First order	Total
deposit density	0.01 (0.00 to 0.01)	0.01 (0.01 to 0.01)
latent heat	0.03 (0.03 to 0.04)	0.03 (0.03 to 0.04)
pipe temperature	0.00 (0.00 to 0.01)	0.00 (0.00 to 0.00)
heat-coupling fraction	0.29 (0.27 to 0.31)	0.29 (0.28 to 0.30)
non-condensable factor	0.00 (0.00 to 0.00)	0.00 (0.00 to 0.00)
contact length	0.00 (0.00 to 0.00)	0.00 (0.00 to 0.00)
steam temperature	0.00 (0.00 to 0.00)	0.00 (0.00 to 0.00)
condensation multiplier	0.00 (0.00 to 0.00)	0.00 (0.00 to 0.00)
retained film	0.00 (0.00 to 0.00)	0.00 (0.00 to 0.00)

Table S4. Sobol indices of the log10 melting steam per meter, retained film lognormal (N = 8,192 base samples).

S4. Correlated inputs

ρ (T0, P_gen)	ρ (thickness, melt point)	P(pack ≤ 55 °C) (95% interval)	Median range (m)	P(range ≥ 100 m)
0.0	0.0	99.1% (98.7–99.4%)	133.1	64%
0.6	0.0	97.4% (96.7–97.9%)	132.9	64%
−0.6	0.0	99.8% (99.6–99.9%)	133.4	64%
0.0	0.6	99.1% (98.7–99.4%)	133.8	63%
0.0	−0.6	99.1% (98.7–99.4%)	133.6	65%
0.6	0.6	97.4% (96.7–97.9%)	133.5	63%

Table S5. Gaussian-copula correlation of starting temperature with heat generation and of deposit thickness with melt point (3,000 draws).

S5. Dynamic film and design search

Case	Median range (m)	Median film (mm)	P(range ≥ 100 m)
baseline	17.7	0.35	0.3%
viscosity x0.25	23.9	0.24	1.4%
viscosity x4	13.1	0.50	0.1%
drain path x0.25	23.9	0.24	1.4%
drain path x4	13.1	0.50	0.1%
residual film 0.02 mm	17.1	0.36	0.2%
residual film 0.05 mm	16.2	0.39	0.2%
steam shear 0.1 Pa	18.0	0.34	0.3%
steam shear 1 Pa	20.5	0.29	0.7%
steam shear 10 Pa	32.8	0.16	4.2%
wall near the invert (sin 0.2)	13.5	0.48	0.1%
wall at the spring line (sin 1.0)	19.2	0.32	0.5%

Table S6. Dynamic film: sensitivity of the median range and of the final film to the drainage assumptions.

Jacket	Aerogel (mm)	Mission (min)	Setpoint (m/s)	Hull OD (mm)	P(≤ 55 °C)	Range (m)	Steam (kg/m)	Energy (Wh)
evacuated	40	60	0.20	216	97.0%	177.7	1.73	69
evacuated	35	60	0.20	206	96.5%	177.6	1.73	69
evacuated	30	60	0.20	196	95.7%	177.3	1.73	69
atmospheric	40	45	0.20	216	99.1%	133.7	1.73	53
atmospheric	35	45	0.20	206	98.7%	133.6	1.73	53
evacuated	25	45	0.20	186	99.1%	133.6	1.73	52
evacuated	30	45	0.20	196	99.3%	133.6	1.73	52
evacuated	35	45	0.20	206	99.5%	133.6	1.73	52
evacuated	40	45	0.20	216	99.6%	133.6	1.73	52
evacuated	20	45	0.20	176	98.7%	133.5	1.73	52
evacuated	15	45	0.20	166	96.7%	133.1	1.73	52

Jacket	Aerogel (mm)	Mission (min)	Setpoint (m/s)	Hull OD (mm)	P(≤ 55 °C)	Range (m)	Steam (kg/m)	Energy (Wh)
evacuated	25	30	0.20	186	100.0%	89.2	1.73	35
atmospheric	20	30	0.20	176	99.8%	89.1	1.73	35
evacuated	15	30	0.20	166	99.8%	89.1	1.73	35
evacuated	20	30	0.20	176	100.0%	89.1	1.73	35
evacuated	10	30	0.20	156	99.2%	89.1	1.73	35

Table S7. Pareto-optimal feasible designs for instant removal (up to 20, by range).

Jacket	Aerogel (mm)	Mission (min)	Setpoint (m/s)	Hull OD (mm)	P(≤ 55 °C)	Range (m)	Steam (kg/m)	Energy (Wh)
evacuated	40	60	0.05	216	97.0%	23.6	1.73	69
evacuated	40	60	0.10	216	97.0%	23.6	1.73	69
evacuated	40	60	0.20	216	97.0%	23.6	1.73	69
evacuated	35	60	0.05	206	96.5%	23.5	1.73	69
evacuated	35	60	0.10	206	96.5%	23.5	1.73	69
evacuated	35	60	0.20	206	96.5%	23.5	1.73	69
evacuated	30	60	0.05	196	95.7%	23.4	1.73	69
evacuated	30	60	0.10	196	95.7%	23.4	1.73	69
evacuated	30	60	0.20	196	95.7%	23.4	1.73	69
evacuated	35	45	0.05	206	99.5%	17.8	1.73	52
evacuated	35	45	0.10	206	99.5%	17.8	1.73	52
evacuated	35	45	0.20	206	99.5%	17.8	1.73	52
evacuated	40	45	0.05	216	99.6%	17.8	1.73	52
evacuated	40	45	0.10	216	99.6%	17.8	1.73	52
evacuated	40	45	0.20	216	99.6%	17.8	1.73	52
evacuated	25	45	0.05	186	99.1%	17.7	1.73	52
evacuated	25	45	0.10	186	99.1%	17.7	1.73	52
evacuated	25	45	0.20	186	99.1%	17.7	1.73	52
evacuated	30	45	0.05	196	99.3%	17.7	1.73	52
evacuated	30	45	0.10	196	99.3%	17.7	1.73	52

Table S7. Pareto-optimal feasible designs for dynamic film (up to 20, by range).

NPS (in)	0	10	20	30	40	50
8	18%	4%	1%	0%	0%	0%
10	96%	87%	62%	19%	2%	0%
12	100%	100%	99%	97%	58%	4%
14	100%	100%	99%	97%	58%	4%
16	100%	100%	99%	97%	58%	4%

Table S8. Feasible envelope: P(pack ≤ 55 °C, 45 min) by nominal pipe size (rows) and pipe temperature (columns, °C), 400 draws per cell.

S6. Pipe length and logistics (all cases)

Mission	Melt removal	L (m)	P(one)	Dep. median	Dep. p95	Active (h)	Total (h)	Range (m)
45 min atmospheric	ablation	0.3	100%	1	1	0.00	11.5 (8.3–14.7)	133.6
45 min atmospheric	ablation	10	100%	1	1	0.06	11.6 (8.4–14.8)	133.6
45 min atmospheric	ablation	100	65%	1	3	0.56	12.9 (9.1–18.4)	133.6
45 min atmospheric	ablation	300	0%	3	8	1.68	17.1 (12.0–31.5)	133.6
45 min atmospheric	ablation	600	0%	5	16	3.36	22.6 (14.9–51.9)	133.6
45 min atmospheric	ablation	1,000	0%	8	26	5.61	30.6 (18.2–79.6)	133.6
45 min atmospheric	retained 0.5 mm	0.3	100%	1	1	0.02	11.5 (8.3–14.7)	12.5
45 min atmospheric	retained 0.5 mm	10	62%	1	3	0.60	13.1 (9.1–18.5)	12.5
45 min atmospheric	retained 0.5 mm	100	0%	9	27	6.00	32.1 (16.8–81.2)	12.5
45 min atmospheric	retained 0.5 mm	300	0%	25	79	18.01	75.3 (30.6–224.9)	12.5

Mission	Melt removal	L (m)	P(one)	Dep. median	Dep. p95	Active (h)	Total (h)	Range (m)
45 min atmospheric	retained 0.5 mm	600	0%	49	158	36.02	140.0 (50.0–438.8)	12.5
45 min atmospheric	retained 0.5 mm	1,000	0%	81	263	60.03	227.1 (76.3–723.3)	12.5
45 min atmospheric	no drainage	0.3	97%	1	1	0.10	11.7 (8.5–15.1)	2.3
45 min atmospheric	no drainage	10	9%	5	28	3.30	22.7 (11.6–82.2)	2.3
45 min atmospheric	no drainage	100	0%	45	274	32.99	130.5 (29.2–725.2)	2.3
45 min atmospheric	no drainage	300	0%	133	822	98.96	369.0 (66.2–2,154.9)	2.3
45 min atmospheric	no drainage	600	0%	265	1,643	197.92	727.6 (121.5–4,301.2)	2.3
45 min atmospheric	no drainage	1,000	0%	442	2,738	329.86	1,206.4 (195.9–7,162.3)	2.3
90 min evacuated	ablation	0.3	100%	1	1	0.00	11.5 (8.3–14.7)	255.2
90 min evacuated	ablation	10	100%	1	1	0.06	11.6 (8.4–14.8)	255.2
90 min evacuated	ablation	100	89%	1	2	0.56	12.4 (9.0–16.2)	255.2
90 min evacuated	ablation	300	41%	2	5	1.68	14.9 (10.1–24.5)	255.2
90 min evacuated	ablation	600	0%	3	9	3.37	18.8 (13.0–38.2)	255.2
90 min evacuated	ablation	1,000	0%	4	14	5.61	23.9 (15.0–57.0)	255.2
90 min evacuated	retained 0.5 mm	0.3	100%	1	1	0.02	11.5 (8.3–14.7)	23.8
90 min evacuated	retained 0.5 mm	10	89%	1	2	0.60	12.4 (9.0–16.3)	23.8
90 min evacuated	retained 0.5 mm	100	2%	5	14	6.02	24.9 (14.4–57.8)	23.8
90 min evacuated	retained 0.5 mm	300	0%	13	42	18.05	53.4 (23.9–151.6)	23.8
90 min evacuated	retained 0.5 mm	600	0%	26	84	36.10	95.9 (37.1–292.9)	23.8
90 min evacuated	retained 0.5 mm	1,000	0%	43	140	60.17	152.5 (54.9–480.9)	23.8
90 min evacuated	no drainage	0.3	99%	1	1	0.10	11.7 (8.4–14.9)	4.3
90 min evacuated	no drainage	10	23%	3	15	3.29	18.6 (10.6–58.7)	4.3
90 min evacuated	no drainage	100	0%	24	146	32.89	88.6 (23.2–498.1)	4.3
90 min evacuated	no drainage	300	0%	70	438	98.66	245.1 (47.9–1,473.8)	4.3
90 min evacuated	no drainage	600	0%	138	875	197.31	479.7 (84.6–2,936.8)	4.3
90 min evacuated	no drainage	1,000	0%	230	1,458	328.86	792.3 (134.8–4,891.9)	4.3
45 min atmospheric	dynamic film	0.3	100%	1	1	0.01	11.5 (8.3–14.7)	17.7
45 min atmospheric	dynamic film	10	80%	1	2	0.42	12.4 (8.9–16.6)	17.7
45 min atmospheric	dynamic film	100	0%	6	18	4.22	25.9 (14.8–57.3)	17.7
45 min atmospheric	dynamic film	300	0%	17	52	12.66	56.7 (25.0–150.5)	17.7
45 min atmospheric	dynamic film	600	0%	34	104	25.33	102.8 (39.1–291.6)	17.7
45 min atmospheric	dynamic film	1,000	0%	57	172	42.21	164.4 (58.2–479.6)	17.7
90 min evacuated	dynamic film	0.3	100%	1	1	0.01	11.5 (8.3–14.7)	33.7
90 min evacuated	dynamic film	10	96%	1	1	0.42	12.1 (8.8–15.5)	33.7
90 min evacuated	dynamic film	100	5%	3	10	4.22	20.8 (12.8–41.7)	33.7
90 min evacuated	dynamic film	300	0%	9	28	12.66	41.0 (19.9–104.1)	33.7
90 min evacuated	dynamic film	600	0%	18	56	25.33	71.2 (29.8–198.0)	33.7
90 min evacuated	dynamic film	1,000	0%	30	93	42.21	111.7 (42.6–321.9)	33.7

Table S9. Deployments, active time and total intervention time by mission, melt-removal case and pipe length (median; total with 5th–95th percentile).

S7. Pipe size and temperature (detail)

NPS (in)	Bore (mm)	Hull OD (mm)	Pack radius (mm)	Aerogel (mm)	P atm.	P evac.	h scale	Steam scale
4	102.3	–	25	does not fit	–	–	1.31	0.34
6	154.1	65	25	does not fit	–	–	1.19	0.51
8	202.7	114	25	24	0.0% (0.0–0.5%)	5.1% (3.8–6.9%)	1.11	0.67
10	254.5	166	35	40	42.2% (38.9–45.7%)	50.0% (46.5–53.5%)	1.05	0.83
12	304.8	216	60	40	99.2% (98.4–99.7%)	99.4% (98.5–99.7%)	1.00	1.00
14	336.5	248	60	40	99.2% (98.4–99.7%)	99.4% (98.5–99.7%)	0.98	1.10
16	387.3	298	60	40	99.2% (98.4–99.7%)	99.4% (98.5–99.7%)	0.94	1.27

Table S10. Fit and thermal margin by nominal pipe size (800 draws; Wilson interval). h scale: condensation coefficient relative to 12 in; steam scale: melting steam per meter relative to 12 in.

Temperature (°C)	P atm. (95% interval)	P evac. (95% interval)	Median pack T at 45 min (°C)	Melt time (s)	Range (m)
0	100.0% (99.5–100.0%)	100.0% (99.5–100.0%)	14.3	13.1	111
10	100.0% (99.5–100.0%)	100.0% (99.5–100.0%)	24.2	12.3	119
25	98.8% (97.7–99.3%)	99.9% (99.3–100.0%)	39.1	10.9	133
35	84.5% (81.8–86.8%)	89.8% (87.5–91.7%)	49.0	10.1	139
45	21.9% (19.1–24.9%)	35.9% (32.6–39.3%)	58.9	9.3	102
55	0.4% (0.1–1.1%)	0.1% (0.0–0.7%)	68.7	8.4	0

Table S11. Pipe / fluid temperature (12 in design, 40 mm aerogel; 800 draws): probability, median pack temperature at 45 min (atmospheric), median melt time and range.

S8. Bypass: peak speed by pocket volume

V_g (m³)	0	5	10	20	40	80	160	320
0.3	97%	97%	96%	95%	92%	88%	82%	72%
1.0	98%	98%	98%	98%	97%	95%	92%	87%
3.0	99%	99%	99%	98%	98%	98%	96%	94%
10.0	99%	99%	99%	99%	99%	99%	98%	98%

Table S12. P(peak speed > 1 m/s) against bypass area (cm²; columns) for four gas-pocket volumes (rows, m³); other inputs sampled.

V_g (m³)	20 cm²	80 cm²	320 cm²
0.3	76	19	5
1.0	254	64	16
3.0	762	190	48
10.0	2,538	635	159

Table S13. Median time for a fully open bypass to halve the breakthrough differential (ms), tool held at rest, against pocket volume and bypass area.

Q_s (m³/s at upstream conditions)	Bypass area (cm²)
0.01	1.6
0.03	4.7
0.10	15.6
0.30	46.9
1.00	156.3

Table S14. Bypass flow area that holds a 0.5 bar differential at 18 bar upstream (assumed methane-like gas, C_d = 0.7) for a supply flow Q_s.

S9. Maintenance lattice (first suite)

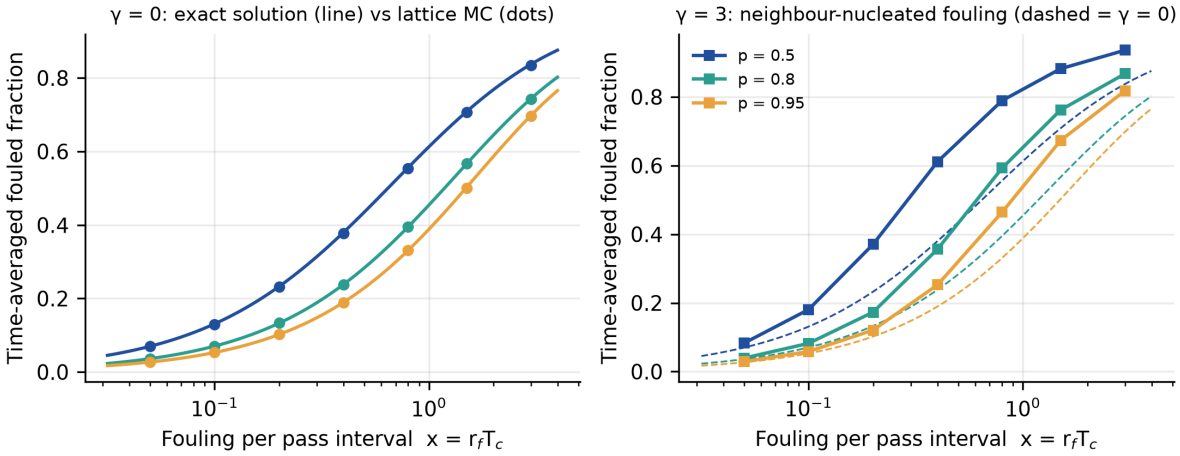


Fig. S4. Time-averaged fouled fraction against fouling per pass interval for three clearing probabilities: non-interacting case (left) and neighbor-nucleated fouling, γ = 3 (right).

For γ=0 the lattice Monte Carlo agrees with the exact solution within 0.26 percentage points. Neighbor-nucleated fouling raises the fouled fraction substantially: for x=0.8 and p=0.8 it rises from 0.40 to 0.59, so a cleaning interval that suffices for independent fouling may fail when deposits nucleate on existing deposits. The lattice is not calibrated to any deposit growth rate and is reported as a planning illustration only.

S10. Automated checks

tests/test_all.py runs 62 checks. First suite: unchanged (sha256) and reproduces its published results; the maintenance-lattice formula. Extension: the extended thermal, melt and surge functions equal the first suite when the added mechanisms are off; monotonic range with retained film; the no-drainage limit; the series-resistance flux; the $D^{-1/4}$ scaling of the condensation coefficient; zero and choked orifice flow; gas-fraction bounds; monotonic bypass benefit; venting-time scaling; proportionality of the holding area to the supply flow; the 12 in hull; the fit in 4, 6 and 8 in pipes; the wall-heating energy balance; mesh convergence; the 270 m and 540 m caps. Revision: the enthalpy solver against the exact Neumann solution; the ablation formula and the no-drainage bias against the solver; convergence of the ablation solution; the conduction solver against an independent stiff-ODE integration; the Chato to Nusselt ratio; restoration of the swapped condensation coefficient; the dynamic film in its two limits; the series-resistance identity and the Π_f threshold; IF97 saturation pressures; the Sobol estimator on a function with known indices and the sums of the computed first-order indices; the Pareto mask; the constraints of the design search; the Class 150 limit; the speed-cap accounting; and the sign of the correlation effect.

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